

GHOUSE MOHIUDDIN MOHAMMED

Backend & AI Engineering | Python · TypeScript · Postgres · AWS | LLM Systems · React

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SUMMARY

Backend-leaning full stack engineer with **8 months of production experience at a startup**, where I owned two systems end to end with minimal supervision. Build with LLMs hands-on: a RAG system with a four-strategy retrieval comparison and evaluation harness, plus **8 months working agenticallly** in production with Claude Code and MCP tool calling. Python, TypeScript, Postgres, AWS. MS Information Systems, graduating December 2026, Toronto-based.

WORK EXPERIENCE

Full Stack Developer Co-op

Jan 2026 – Aug 2026

Apex Mobile Media (startup)

Toronto, ON

- **Owned two production systems end to end with minimal supervision**, in a small team with no dedicated platform or infra support: scoped the problems myself, made the architecture calls, deployed them, and handled production issues when they broke.
- Built backend services and REST APIs (**Python/FastAPI**) on **AWS** (EC2, RDS, S3, ALB, ECR, CloudWatch, IAM) with Docker, backed by **PostgreSQL** and Redshift: schema design, query construction and optimization.
- Shipped the frontend end to end where features required it (**React/TypeScript**): **about 60** daily users, **30+ modules**, **10-role** access control.
- Re-architected a nightly data migration independently after benchmarking it at **24 hours**: traced the cost to redundant full reloads and rebuilt it as a two-tier hot/cold design with hash-based change detection, cutting runtime to **under 15 minutes**.
- Cut a core query endpoint from **411s to 28ms** via materialized-view rollups and indexing strategy; ran **34M+ row** backfills against a live system with no user-facing downtime.
- Designed ETL pipelines ingesting **12+ heterogeneous sources** into a **400M+ record** warehouse, with retry, reconciliation and monitoring so failures surfaced before users saw them.
- Worked directly with finance, sales and account management to scope requirements and ship features end to end, cutting month-end reconciliation effort by **over 80%**.

Graduate Teaching Assistant

Sep 2026 – Present

Northeastern University

Toronto, ON

- Support instruction for **Program Structures and Algorithms** (INFO 6205) and two graduate data engineering courses: review student code and run office hours on data structures, complexity analysis and system design.

LLM & AI SYSTEMS

Flight Refund RAG Assistant

github.com/ghousee/flight-refund-rag

- Retrieval-augmented generation system over airline and regulatory policy documents. Compared **four retrieval strategies** (vector, hybrid, reranked, metadata-filtered) against a hand-authored golden set, measuring recall@5, MRR and RAGAS faithfulness. Python, LangChain, PostgreSQL/pgvector, Docker.
- Diagnosed **AI-specific failure modes** rather than assuming the pipeline worked: found cases where retrieval returned the correct passages but the model still mis-synthesized the answer, which is invisible without an evaluation harness separating retrieval quality from generation quality.

Agentic development in production

Apex Mobile Media

- Used **Claude Code and MCP servers** (live API documentation as tool calls) as my primary implementation workflow for 8 months on systems serving real users, while owning architecture, code review and testing. Practical experience with prompt design, tool calling, and where these systems fail: bad output that looks correct, latency, and dropped calls.

AI Model Reasoning Benchmark

Manipal Institute of Technology

- Four-month research project developing a binary-input approach to benchmark reasoning capability in AI models. Recommended by faculty for the Manipal Research Colloquium 2023.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Java

Backend: FastAPI, REST API design, data modeling, asynchronous processing, retry and idempotency, monitoring and alerting, production debugging

Frontend: React, TypeScript, component architecture, state management

Databases: PostgreSQL, pgvector, Amazon Redshift, DuckDB; schema design, query optimization, indexing

Cloud: AWS (EC2, RDS, S3, ALB, ECR, CloudWatch, IAM), Docker, Linux, Git, CI/CD

AI: RAG patterns, LangChain, embeddings and vector search, prompt design, tool calling (MCP), evaluation harness design

EDUCATION

MS Information Systems (College of Engineering)

Sep 2024 – Dec 2026

Northeastern University, Toronto, ON

GPA: 3.84 / 4.0

B.Tech Information Technology

Jul 2019 – Jul 2023

Manipal Institute of Technology, Manipal, India